

PIOTR SYNAK, PhD

RESEARCH AND DEVELOPMENT DIRECTOR | AI/ML EXPERT



SUMMARY

AI-driven Research and Development Director with over **7 years of leadership** experience and **20 years in software engineering**. **Expert in AI/ML** systems, including anomaly detection and data mining, with a strong track record of project management in cross-functional, international environments. Co-inventor of multiple **database-related patents**, and experienced in aligning cutting-edge technology with business strategy.

CONTACT & INFO



Email:
piotr.d.synak@gmail.com



Address:
Winterthur, Switzerland



Website
<https://www.researchgate.net/profile/Piotr-Synak>



Personal
• Polish/Swiss citizenship

SKILLS

- AI/ML System Design & Deployment
- Agile & Project Management (Scrum, Kanban)
- Cross-functional Team Leadership
- Software Architecture & Engineering
- AI-based Anomaly Detection
- Grant Writing & R&D Strategy
- CI/CD, Quality Assurance & Automation
- Innovation and Creativity
- Cybersecurity Applications

EDUCATION

xMBA Agile Business, Startup Skills

SAQ Swiss Association for Quality
2025

PhD/COMPUTER SCIENCE

Institute of Computer Science, Polish
Academy of Sciences, Poland

M.Sc./MATHEMATICS

Faculty of Mathematics, Informatics, and
Mechanics, Warsaw University, Poland

SELECTED ACHIEVEMENTS

- DeepSeas (2022-2024) – Architected **AI/ML-driven** volumetric anomaly-detection platform, cutting alert latency by 80 % and scaling to 10 B+ logs/day on AWS.
- Security On-Demand (2017-2022) – Created an **approximate query database engine**; reduced large-scale threat-hunting queries from hours to seconds.
- Infobright (2005-2017) – Co-founded and led design of **C++ columnar database** adopted by Fortune 500 analytics workloads.

EXPERIENCE

AI CONSULTANT AND RESEARCH ENGINEER

Portfolio Career / Switzerland / 2025 –

Guided a pre-seed-stage startup in architecting an **LLM-powered** platform that transforms internal documentation into rich knowledge graphs, dramatically accelerating new-hire onboarding.

- NLP**: Custom spaCy pipelines for entity extraction and relation linking
- LLM stack**: OpenAI (GPT-5 family), Anthropic (Claude 4), Google Gemini
- Embeddings**: Sentence Transformers, OpenAI Embeddings

Designed and implemented a proprietary local Retrieval-Augmented Generation (RAG) pipeline and agentic architecture for a new startup.

- Agentic orchestration**: Applied multi-step reasoning using custom agents built with LangGraph and LlamaIndex.
- Demonstrates deep expertise in designing AI-enabled knowledge systems and agent architectures for real-world enterprise scenarios.

DIRECTOR / MANAGING DIRECTOR OF RESEARCH AND DEVELOPMENT

Security On-Demand (acquired and rebranded as DeepSeas in 2022) /
USA-Poland-Switzerland / 2017 – 2024

Chief architect and R&D lead at Security On-Demand/DeepSeas, a globally distributed managed security services provider delivering 24/7 threat monitoring and response. Developed an **AI-driven** Approximate Query database and anomaly-detection engine for real-time analysis of billions of security logs.

- Led **AI/ML** development of a **volumetric anomaly detection** system for cybersecurity, processing billions of logs daily.
- Architected and designed the **Approximate Query (AQ) database engine** for cybersecurity log analytics and anomaly detection.
- Designed an **AI/ML-driven data analysis** system for high-throughput, real-time log processing using approximation queries.
- Managed the end-to-end roadmap for AQ database development using AWS cloud-native architecture.
- Managed international R&D teams and Agile processes, and implemented automated **CI/CD pipelines** using Jenkins.

• TECHNICAL SKILLS

- Platforms:
Windows, Linux
- Languages:
C / C++, Python, SQL, Java
- Databases:
Infobright, Postgresql, MySql, SQLServer
- AI/ML Frameworks:
PyTorch, LangGraph, LlamaIndex
- Tools:
Cursor, KiloCode, Windsurf, Bolt.new, svn, GitHub, Azure DevOps (ADO), RallyDev, Jenkins, Confluence, Visual Studio, Eclipse

SCIENTIFIC AREAS

- Artificial Intelligence & Machine Learning
- Rough Sets
- Database Engine Design
- Temporal Data Analysis
- Multi-agent Systems
- Data Mining
- Soft Computing

INTERESTS

- Mastering sushi
- Ferroequinologist
- Japanese model trains
- Butoh
- Judo
- Football
- Hiking
- Photography

LANGUAGES

- Polish — mother tongue
- English — fluent
- German — intermediate

EXPERIENCE - CONTINUED -

CO-FOUNDER, CHIEF ARCHITECT AND LEAD DESIGNER

Infobright / Canada-Poland / 2005 – 2017

Chief software architect, lead designer, and developer of Infobright — an **AI-powered columnar database** engine written in C++ with Postgresql/MySQL frontend (Windows, Linux) for high-speed processing of large amounts of data. Designed and implemented core data structures and algorithms for query execution, including query execution plans, rough queries, correlated sub-queries, joins, and aggregation.

- Participated in key architectural discussions and decisions regarding query execution engines, parallelization, memory optimization, scaling (out and up), partitioning, high availability, and open-source transition.
- Served as Scrum Master for the database design group, leading a cross-functional team to develop core query engine functionality and implement Agile methodology.
- Supported sales engineers in resolving issues/requests filed by customers.

LEAD DESIGNER AND DEVELOPER

Cognitron / Canada-Poland / 1997 – 2004

Led the design and development of RS Explorer — an AI/ML system for data exploration and mining, implementing ideas of the theory of rough sets.

ASSISTANT / ADJUNCT PROFESSOR

Polish-Japanese Academy of Information Technology / Poland / 1996 – 2009

Conducted lectures and labs on Data Mining, AI, Algorithms, C++, and Java.

PATENTS AND RESEARCH

- Participated in over **15 research projects** and EU grants.
- Presented at several international conferences.
- Published more than **60 research papers**, including book chapters, journal articles, and conference proceedings.
- Co-inventor of **three patents** and **one patent application** in the area of databases: US8700579, US8838593, US11301467, US20150088807A1.
- Co-author of AI/ML systems: RSES (www.mimuw.edu.pl/~szczuka/rses) and Rosetta (bioinf.icm.uu.se/rosetta).
- xMBA Startup-Coach and Agile Business Expert (SAQ-certified)
- Visiting Assistant, University of North Carolina at Charlotte, USA
- Awarded the Japan International Cooperation Agency's scholarship for multimedia training at Saitama University, Saitama, Japan.
- Visiting assistant, Norwegian University of Science and Technology, Trondheim, Norway.
- Awarded a Tempus scholarship for software engineering training at Italsiel/Finsiel, Rome, Italy.

REFERENCES

RICK GLICK

Former CTO
Teradata

P: +1 (760) 224-4169
E: rick.glick@gmail.com

JOEL HOLLAND

CTO
DeepSeas

P: +1 (858) 405-5635
E: joel@hollandfamily.online

DOMINIK ŚLĘZAK

CEO
QED Software

P: +48 660 495 076
E: dominik.slezak@qed.pl